

Use the following context to answer questions 1-4.

Members of the cross country team are preparing for a 3-mile race. The average time for the team is 21.25 minutes. The fastest and slowest times varied from the average by 3.1 minutes.

1. Define the variable.

Let x = number of minutes

2. Write an absolute value inequality that represents the context.

$$|x - 21.25| \leq 3.1$$

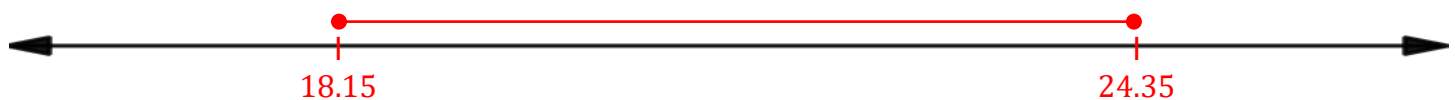
3. Solve the inequality. Write the answer as a compound inequality.

$$\begin{array}{r} x - 21.25 \geq -3.1 \\ +21.25 \quad +21.25 \\ \hline x \geq 18.15 \end{array}$$

$$\begin{array}{r} x - 21.25 \leq 3.1 \\ +21.25 \quad +21.25 \\ \hline x \leq 24.35 \end{array}$$

$$18.15 \leq x \leq 24.35$$

4. Graph the solution on the number line.



Use the following context to answer questions 5-6.

Noah is a member of the high school band. Each day he spends an average of 62 minutes practicing for an upcoming band concert. He spent less time practicing on school days, but more time on the weekends. Each of his daily practice times was within 29 minutes of his average.

5. Write an absolute value inequality that represents m , the number of minutes Noah spends practicing for the band concert on a given day.

$$|m - 62| \leq 29$$

6. Solve the inequality.

$$-29 \leq m - 62 \leq 29$$

$$\begin{array}{ccc} +62 & +62 & +62 \\ \hline \end{array}$$

$$33 \leq m \leq 91$$

between 33 & 91 minutes

Use the following context to answer questions 7-10.

Payton made a personal goal of earning \$145 per week at her after-school job. Each week, she has been averaging within \$19.23 of her goal.

7. Define the variable.

Let x = the amount of her weekly paycheck.

8. Write an absolute value inequality that represents the context.

$$|x - 145| \leq 19.23$$

9. Solve the inequality. Write the answer using set-builder notation.

$$-19.23 \leq x - 145 \leq 19.23$$

$$\begin{array}{ccc} +145 & +145 & +145 \\ \hline \end{array}$$

$$125.77 \leq x \leq 164.23$$

$$\{x | 125.77 \leq x \leq 164.23\}$$

10. Graph the solution on the number line.

